

MediBridge – Use-Case Diagram Explanation

1. System Overview

MediBridge is a healthcare management platform that connects patients, doctors, and administrators through a single web-based system. The system allows patients to find doctors, book appointments, manage medical information and medications, while doctors can manage appointments, patients, medical records and prescriptions. Administrators manage users, doctors, appointments, healthcare resources and other system-level activities.

The system has three primary actors: **Patient, Doctor, and Admin.**

2. Actors

Patient

The Patient is the primary user of MediBridge who uses the platform to access healthcare services.

The patient can:

- Register and log in to the system.
- Search and filter available doctors.
- View doctor profiles and information.
- Book appointments with doctors.
- Cancel or reschedule appointments.
- View medical records and prescriptions.
- Manage medications.
- Track medication intake.
- View their Care Score and Health Timeline.
- Receive and view notifications.
- Submit reviews for doctors.
- Manage their personal profile.
- Access healthcare resources and emergency guidance.

Doctor

The Doctor uses MediBridge to manage their professional activities and interact with patients.

The doctor can:

- Log in to the system.

- Manage their professional profile.
- Manage their availability.
- View scheduled appointments.
- Confirm or cancel appointments.
- View patient information.
- Create and update medical records.
- Add prescriptions for patients.
- View notifications related to appointments and activities.

Admin

The Admin is responsible for managing and maintaining the MediBridge platform.

The admin can:

- Log in to the administration system.
- View the dashboard and system analytics.
- Manage patient accounts.
- Manage doctor accounts.
- Verify doctors.
- Manage appointments.
- Manage medical specializations.
- Manage healthcare resources.
- Manage reported issues.
- View system audit logs.

3. Major Use Cases

Register / Login

Allows patients and authorized users to securely access the MediBridge platform according to their role.

Search & Filter Doctors

Allows patients to search for doctors and filter them based on relevant information such as specialization and availability.

View Doctor Profile

Allows patients to view information about a doctor before deciding to book an appointment.

Book Appointment

Allows a patient to select a doctor and schedule an appointment based on available slots.

Manage Appointments

Allows patients to cancel or reschedule appointments. Doctors can also manage appointment status by confirming or cancelling appointments.

View Medical Records

Allows patients to access their medical history and records maintained through the platform.

View Prescriptions

Allows patients to view prescriptions provided by their doctors.

Manage & Track Medications

Allows patients to manage their medications and record/track their medication intake.

Care Score & Health Timeline

Allows patients to view their healthcare-related progress through the Care Score and Health Timeline features.

Notifications

Allows users to receive important updates and information related to appointments and other activities.

Submit Doctor Review

Allows patients to provide feedback or ratings after interacting with doctors.

Create Medical Record

Allows doctors to create and maintain medical records for their patients.

Add Prescription

Allows doctors to provide prescriptions to patients through the system.

Manage Availability

Allows doctors to specify and manage the time slots during which they are available for appointments.

Admin Dashboard & Analytics

Allows administrators to monitor overall system activity and view important statistics.

Manage and Verify Doctors

Allows administrators to manage doctor accounts and verify doctors before or during their participation in the platform.

Manage Healthcare Resources

Allows administrators to maintain healthcare-related resources available to users.

Manage Reported Issues

Allows administrators to review and manage issues reported by users.

Audit Logs

Allows administrators to view records of important system activities for monitoring and accountability.

4. Relationships Between Actors and Use Cases

The **Patient** is primarily associated with healthcare discovery, appointments, medical information, medications, notifications, reviews and personal profile management.

The **Doctor** is primarily associated with appointment management, patient information, medical records, prescriptions, availability and professional profile management.

The **Admin** is primarily associated with system administration, user management, doctor verification, resources, issue management, analytics and audit logs.

Some use cases have an <<include>> **relationship** where one function supports another. For example, booking an appointment involves finding/searching for a suitable doctor, while doctor management may involve the verification of doctors.

5. Conclusion

The use-case diagram represents the functional requirements of the MediBridge healthcare management system. It shows how the three major actors—**Patient, Doctor and Admin**—interact with the system and its major features. This provides a clear overview of the system's functionality and helps in understanding the responsibilities of each user role.